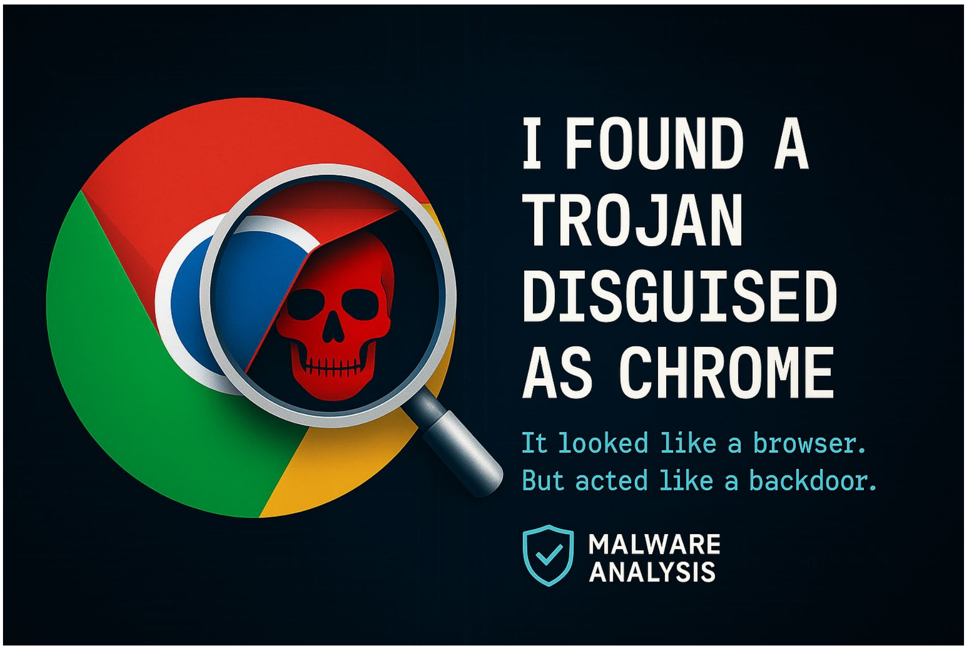


Malware Analysis Report: “BackupWin” Trojan

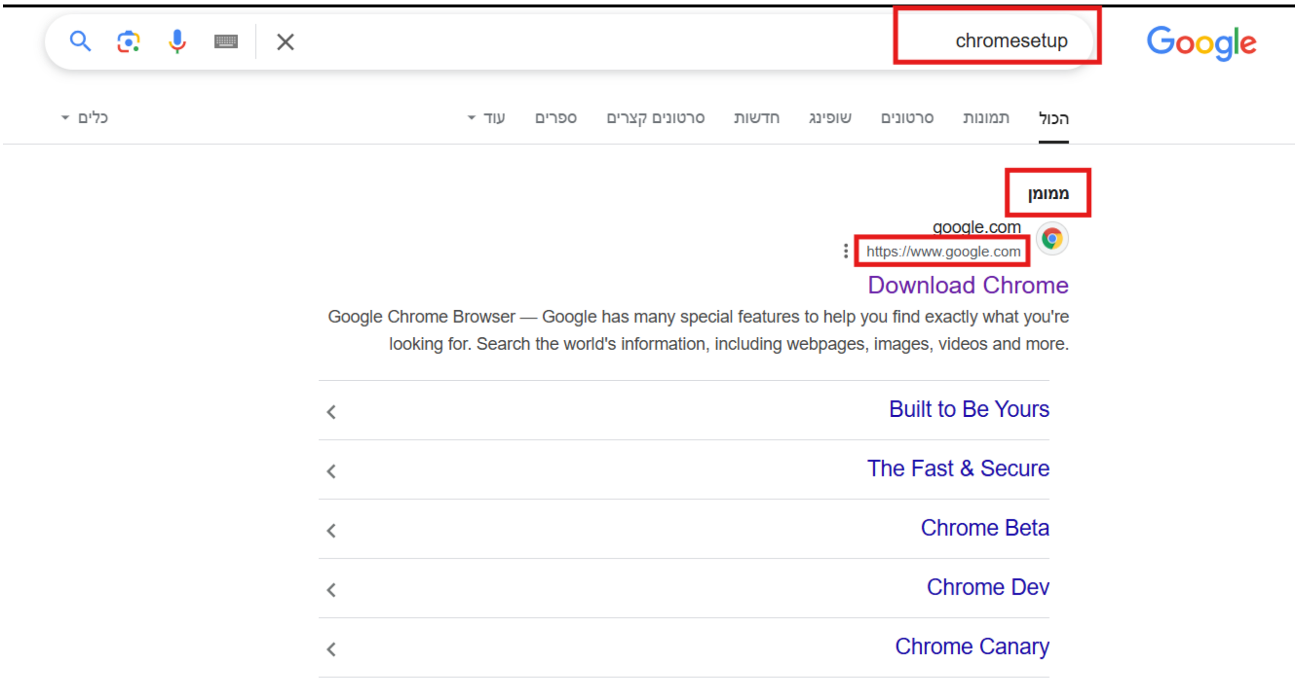
1. Introduction & Discovery Story

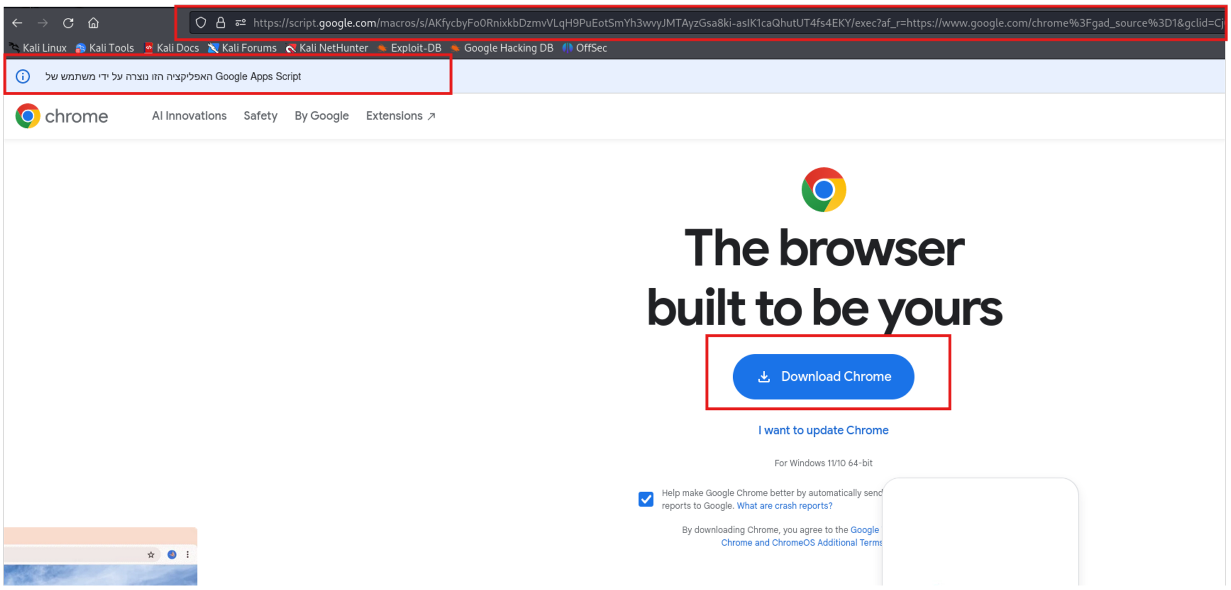
I was searching for **“chromesetup”** on Google and clicked the top **Sponsored** link. The website looked almost identical to the official Chrome download page. However, the URL was a Google Scripts subdomain:

https://script.google.com/macros/s/AKfycbyFo0RnixkbDzmvVLqH9PuEotSmYh3wvyJMTAyzGsa8ki-asIK1caQhutUT4fs4EKY/exec?af_r=...

From there, I clicked a **“Download Chrome”** button, which gave me an EXE called Chrome-64.exe from:

https://chrome.homepagel.com/ (**Warning!** Clicking this link will immediately download the EXE Trojan to your PC, Download it only in secure VM.)





At this point, something felt off:

- Why wasn't it downloading from google.com?
- Why was Google *sponsoring* a link like this?

Instead of installing it, I uploaded the file to **VirusTotal** and ran behavior analysis in a **Kali Linux VM** & **Windows 10 VM** with Analysis Tools.
The results revealed a trojan I'm calling **"BackupWin"** (named after strings and file paths found in the code).

2. Suspected File Info vs Original Chrome Setup Info

	Original Chrome Setup	Suspected Chrome Setup
File Name	**ChromeSetup.exe**	**Chrome-x64.exe**
MD5 Hash	120b91990785664528f3b8829570e055	528607dc316d1c0cd45ac8b72f97068b
SHA-1 Hash	c5a44fdde7bafe53c4c2893b015a8ca3840af219	72c64c02a31736029713270f1dc8f24c0f576d1b
SHA-256 Hash	1e9e2cce86974afadfba1818195ab669f53b5bf4 6165c9b27da1a4c7e83198d7	2ad37c2bc96ab025b2b72601d16ce90904a5afff b3cc37e706308a4700da82d7
File Size	**11MB**	**130MB**
File Type	**PE32 executable (GUI) Intel 80386, for MS Windows**	**PE32 executable (GUI) Intel 80386, for MS Windows**
File Metadata (Exiftool)		
Original File Name	**UpdaterSetup.exe**	**AppApi.dll**
Signature	**Google LLC** Signer information Name: Google LLC E-mail: Not available Signing time: Thursday, March 20, 2025	**LLC Spectr** Signer information Name: LLC Spectr E-mail: Not available Signing time: Wednesday, March 19, 20
Company Name	**Google LLC**	**AppApi**
Product Name	**Installer**	**AppApi**
File Description	**Google Installer**	**AppApi**
File Version	**136.0.7079.0**	**1.0.0.0**

3. VirusTotal Scan

Here's what I discovered from the VirusTotal scan:

- **No antivirus engines flagged it as malicious (possibly a new or obfuscated Trojan).**
- However, in the **Behavior** tab, I noticed suspicious connections to **solarcellled.com** and other domains.
- I visited **https://solarcellled.com/api.php** and found a C# script with clear malware behavior.
- **Please refer to the attachments section below to view the full VirusTotal report.**

4. Suspicious Behavior Observed

I reviewed the **C# script from api.php**. In case it's removed, I've backed it up, (please look on the attachments below):

The script performs the following:

1. Privilege Escalation Attempt

- Tries to **re-launch itself as administrator** using `runas`.

2. Windows Defender Exclusion

- Executes PowerShell:

```
Add-MpPreference -ExclusionPath '%APPDATA%', '%TEMP%'
```

[!] Tells Defender to ignore those folders, where malware usually hides.

3. Encrypted Payload Download

- Downloads an AES-encrypted payload from: `https://solarcellled.com/test.php?uuid=...`
- Decrypts it and saves it as: `%APPDATA%\BackupWin\decrypted.exe`
- Then runs it silently.

4. Persistence via Scheduled Task

Creates a Windows Scheduled Task:

- Name: BackupWinTask
- Action: Run the malware at every login with **highest privileges**.

5. Dynamic Analysis

Main Details

- The Trojan creates folders in hidden directories.
- The Trojan modifies numerous registry entries.
- The Trojan connects to the internet and downloads malicious scripts.
- Overall, The Trojan Downloads, decrypts, and executes a secondary payload; disables Defender; installs a persistence mechanism; hides network activity.

Tools Utilized:

Wireshark, Process Monitor (procmon), Process Explorer (procexp), Regshot, TCPView, FakeNet-NG, Strings.

Suspicous IP Address:

69.62.119.2 (Germany, Hosted by Hostinger) -> https://solarcellled.com + http://srv760637.hstgr.cloud/

69.62.119.17 (Germany, Hosted by Hostinger) -> https://blgzd.com

Malicious Domains:

http://srv760637.hstgr.cloud/

https://blgzd.com

https://solarcellled.com

https://chrome.homepagel.com/

https://script.google.com/macros/s/AKfycbyFo0RnixkbDzmvVLqH9PuEotSmYh3wvyJMTAyzGsa8ki-asIK1caQhutUT4fs4EKY/exec?af_r=...

Directories Created:

C:\Users\Malware\AppData\Local\Temp\.net\Chrome-x64\AV9JX02HGBnwvgGSvHtJPrgR6B2GBbA=

C:\Users\Malware\AppData\Local\Temp_MEI63322

Registry Analysis (via Regshot)

Using Regshot, I captured a snapshot of the registry **before and after running the trojan**.
Cleaned results are available here: https://pastebin.com/Y1f8R9np.
(If you require the full output, please refer to the attachments below.)

This program:

1. Driver Installation: "**WinDivert1.1**"
   - **Registry:** `HKLM\SYSTEM\CurrentControlSet\Services\WinDivert1.1`
   - **File Path:** `C:\Users\Malware\AppData\Local\Temp\_MEI49~1\WinDivert64.sys`

   This driver was loaded into the **kernel**, making it particularly dangerous.
   WinDivert is commonly used for network packet manipulation, enabling traffic redirection, sniffing, or Man-in-the-Middle (MITM) attacks.
   (See supporting screenshots below that shows that it's actually running on the kernel.)

2. Group Policy Manipulation
   - **Registry**: `HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Group Policy\ServiceInstances\{GUID}`
   - **Observation**:
     Unknown GPO service instances were added and deleted — likely to manipulate system policies or bypass restrictions.

3. Network Configuration Manipulation
   - **Registry**: `HKLM\SYSTEM\CurrentControlSet\Services\Tcpip\Parameters\Interfaces\{GUID}`
   - **Observation**:
     The trojan removed DHCP-based IP/DNS settings, possibly to prevent conflict or detection.
     Combined with the WinDivert driver, this suggests **low-level control over network traffic**.

4. New Network Profiles/Signatures
   - **Registry**: `HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\NetworkList\Profiles\{GUID}`
   - **Registry**: `HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\NetworkList\Signatures\Unmanaged\{GUID}`
   - **Observation:**
     Multiple new **network profiles** were registered — potentially spoofed or virtual networks created by the trojan to reroute traffic

5. Suspicious File Extension Hijacking
   - **Registry**: `HKCU\Software\Microsoft\Windows\CurrentVersion\Explorer\FileExts\.com/redirect`
   - **Observation**:

   Malicious hijack of the .com executable file extension was detected.
   It redirected .com files to open with LaunchWinApp.exe, possibly executing malicious code instead of legitimate programs.

Suspicious DLL Files Created

The core logic DLL of the malware is:

- **Filename:** AppApi.dll
- **SHA256:** 2ad37c2bc96ab025b2b72601d16ce90904a5afffb3cc37e706308a4700da82d7
- **Decompiled Source:** https://pastebin.com/60P8uHGb *(via dnSpy)*

This program:

- Downloads malicious code from the internet (api.php)
- **Executes it in memory** using the C# scripting engine (Roslyn)

The malware generated **over 60 additional DLLs**, located in:

- `C:\Users\Malware\AppData\Local\Temp\.net\Chrome-x64`

Attachments

1. **Wireshark traffic:**
    a. GitHub: Click here
2. **api.php source code:**
    a. Pastebin: Click here
    b. GitHub: Click here
3. **AppApi.dll source code:**
    a. Pastebin: Click here
    b. GitHub: Click here
4. **Registry changes that the trojan made:**
    a. Pastebin: Click here
    b. GitHub (Cleaned up version): Click here
    c. GitHub (Full version): Click here
5. **VirusTotal Scan Report**
    a. VirusTotal: Click here

6. Conclusion & Lessons Learned

Even though no antivirus flagged the file, the behavior and script logic clearly show malicious intent. This is a Trojan Downloader designed to bypass detection, run with elevated privileges, and deliver more payloads.

Key Lessons:

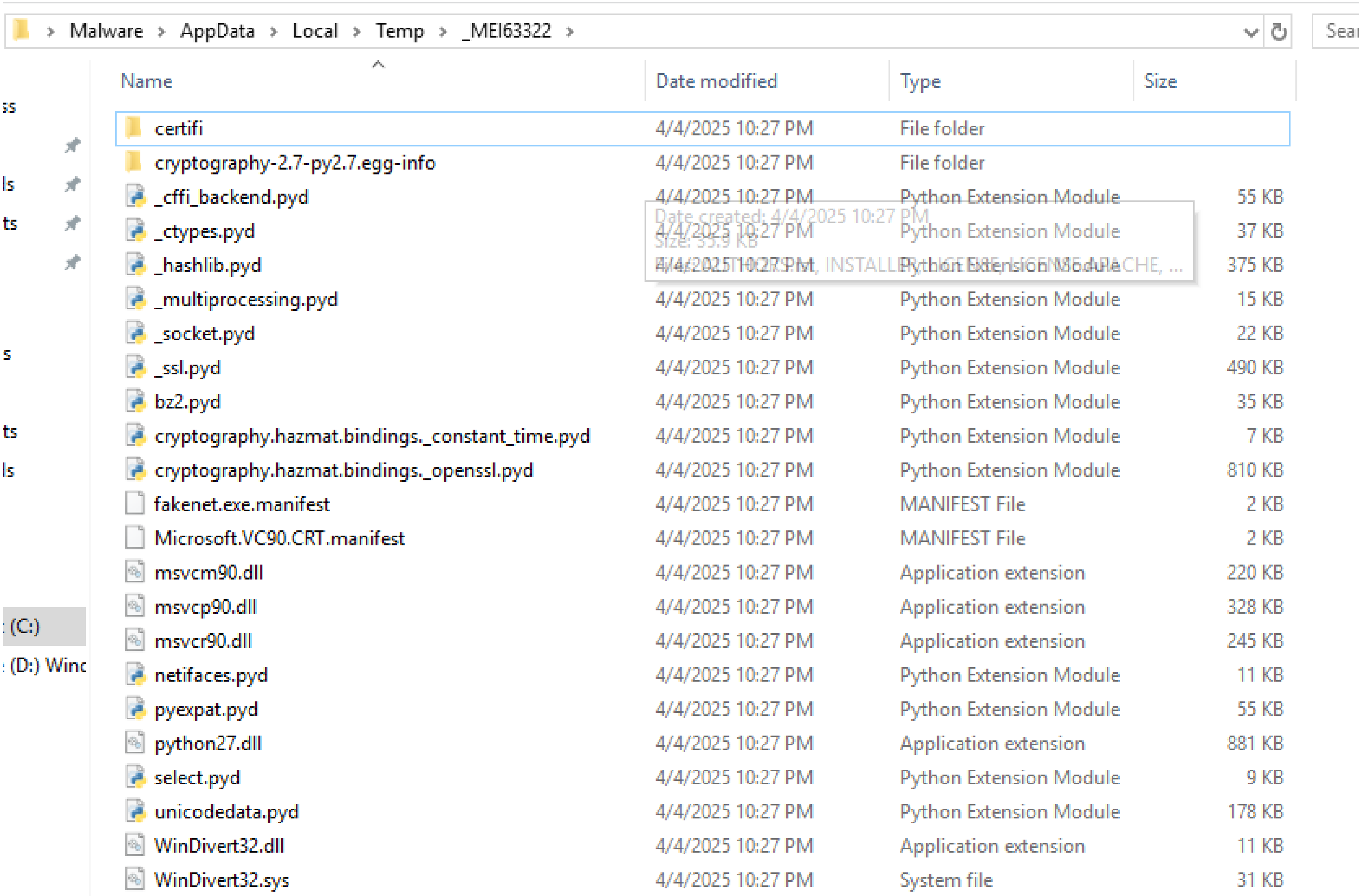
1. No AV detections doesn't mean it's Safe!
2. Always check Behavior tab + external domains
3. Sponsored Ads Can Be Dangerous – even on Google!
4. Verify official domains before downloading software
5. Malware evolves – attackers often bypass signatures
6. Code review reveals truth, even when AV doesn't

7. Additional Notes

1. The attacker frequently changes the URL within the fake Chrome download page, possibly to evade takedowns and blocking lists.
2. The URL https://solarcellled.com/test.php appears to be broken at present.

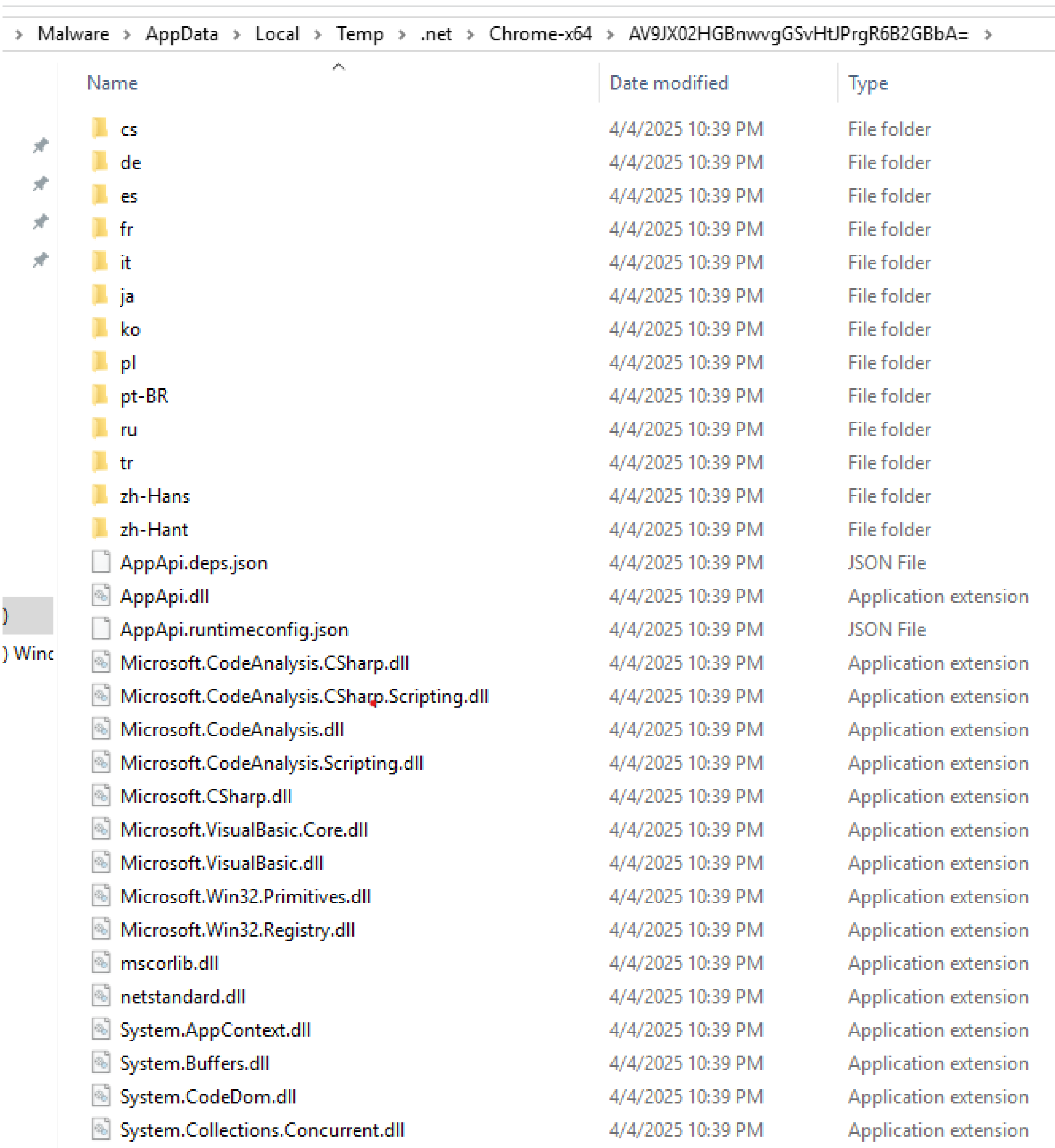
8. Additional Screenshots

1. Screenshot taken from the folder `AppData\Local\Temp\_MEI63322`



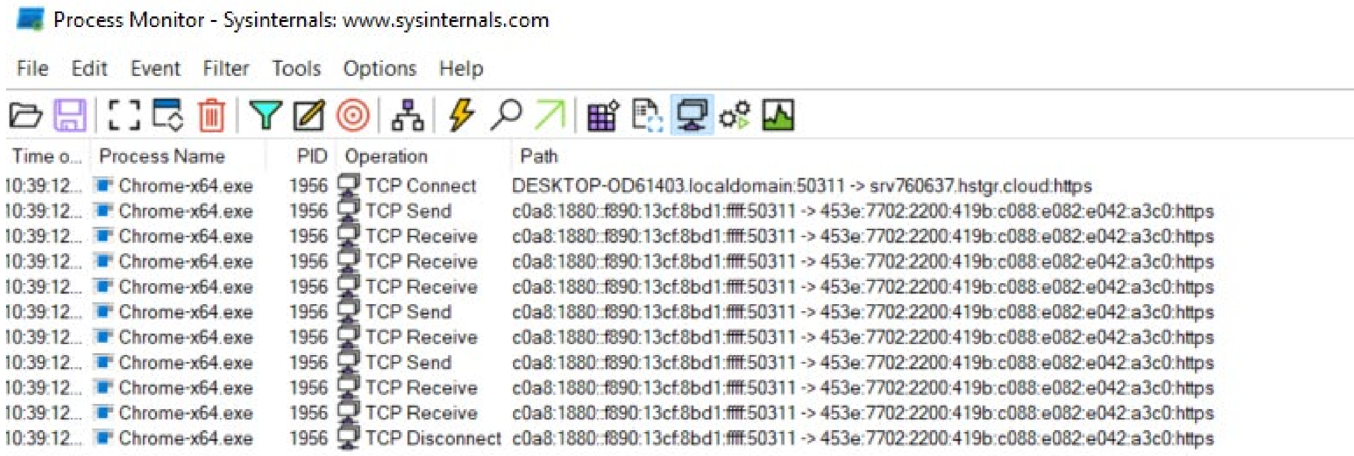
Name	Date modified	Type	Size
certifi	4/4/2025 10:27 PM	File folder	
cryptography-2.7-py2.7.egg-info	4/4/2025 10:27 PM	File folder	
_cffi_backend.pyd	4/4/2025 10:27 PM	Python Extension Module	55 KB
_ctypes.pyd	4/4/2025 10:27 PM	Python Extension Module	37 KB
_hashlib.pyd	4/4/2025 10:27 PM	Python Extension Module	375 KB
_multiprocessing.pyd	4/4/2025 10:27 PM	Python Extension Module	15 KB
_socket.pyd	4/4/2025 10:27 PM	Python Extension Module	22 KB
_ssl.pyd	4/4/2025 10:27 PM	Python Extension Module	490 KB
bz2.pyd	4/4/2025 10:27 PM	Python Extension Module	35 KB
cryptography.hazmat.bindings._constant_time.pyd	4/4/2025 10:27 PM	Python Extension Module	7 KB
cryptography.hazmat.bindings._openssl.pyd	4/4/2025 10:27 PM	Python Extension Module	810 KB
fakenet.exe.manifest	4/4/2025 10:27 PM	MANIFEST File	2 KB
Microsoft.VC90.CRT.manifest	4/4/2025 10:27 PM	MANIFEST File	2 KB
msvcm90.dll	4/4/2025 10:27 PM	Application extension	220 KB
msvcp90.dll	4/4/2025 10:27 PM	Application extension	328 KB
msvcr90.dll	4/4/2025 10:27 PM	Application extension	245 KB
netifaces.pyd	4/4/2025 10:27 PM	Python Extension Module	11 KB
pyexpat.pyd	4/4/2025 10:27 PM	Python Extension Module	55 KB
python27.dll	4/4/2025 10:27 PM	Application extension	881 KB
select.pyd	4/4/2025 10:27 PM	Python Extension Module	9 KB
unicodedata.pyd	4/4/2025 10:27 PM	Python Extension Module	178 KB
WinDivert32.dll	4/4/2025 10:27 PM	Application extension	11 KB
WinDivert32.sys	4/4/2025 10:27 PM	System file	31 KB

2. Screenshot taken from the folder `AppData\Local\Temp\.net\Chrome-x64\AV9JX02HGBnwvgGSvHtJPrgR6B2GBbA=`

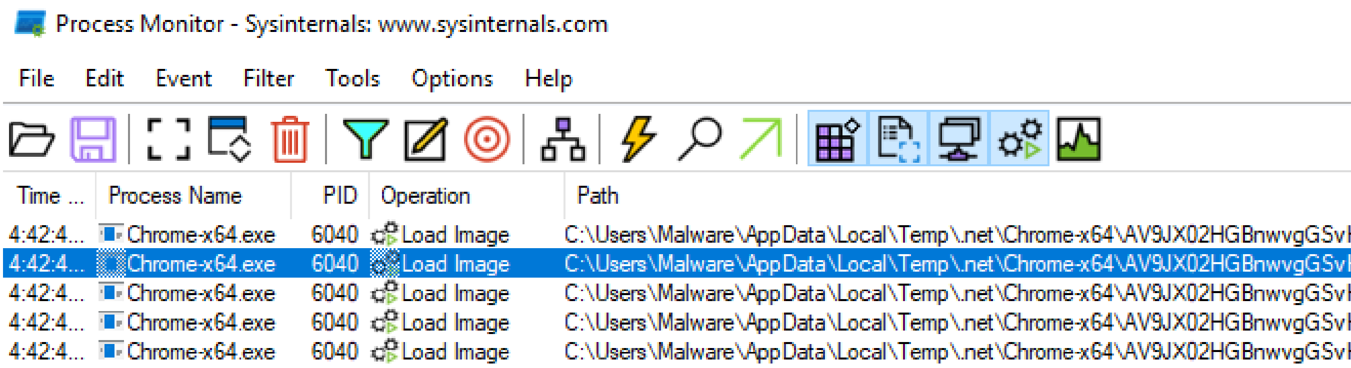

Malware > AppData > Local > Temp > .net > Chrome-x64 > AV9JX02HGBnwvgGSvHtJPrgR6B2GBbA= >

Name	Date modified	Type
cs	4/4/2025 10:39 PM	File folder
de	4/4/2025 10:39 PM	File folder
es	4/4/2025 10:39 PM	File folder
fr	4/4/2025 10:39 PM	File folder
it	4/4/2025 10:39 PM	File folder
ja	4/4/2025 10:39 PM	File folder
ko	4/4/2025 10:39 PM	File folder
pl	4/4/2025 10:39 PM	File folder
pt-BR	4/4/2025 10:39 PM	File folder
ru	4/4/2025 10:39 PM	File folder
tr	4/4/2025 10:39 PM	File folder
zh-Hans	4/4/2025 10:39 PM	File folder
zh-Hant	4/4/2025 10:39 PM	File folder
AppApi.deps.json	4/4/2025 10:39 PM	JSON File
AppApi.dll	4/4/2025 10:39 PM	Application extension
AppApi.runtimeconfig.json	4/4/2025 10:39 PM	JSON File
Microsoft.CodeAnalysis.CSharp.dll	4/4/2025 10:39 PM	Application extension
Microsoft.CodeAnalysis.CSharp.Scripting.dll	4/4/2025 10:39 PM	Application extension
Microsoft.CodeAnalysis.dll	4/4/2025 10:39 PM	Application extension
Microsoft.CodeAnalysis.Scripting.dll	4/4/2025 10:39 PM	Application extension
Microsoft.CSharp.dll	4/4/2025 10:39 PM	Application extension
Microsoft.VisualBasic.Core.dll	4/4/2025 10:39 PM	Application extension
Microsoft.VisualBasic.dll	4/4/2025 10:39 PM	Application extension
Microsoft.Win32.Primitives.dll	4/4/2025 10:39 PM	Application extension
Microsoft.Win32.Registry.dll	4/4/2025 10:39 PM	Application extension
mscorlib.dll	4/4/2025 10:39 PM	Application extension
netstandard.dll	4/4/2025 10:39 PM	Application extension
System.AppContext.dll	4/4/2025 10:39 PM	Application extension
System.Buffers.dll	4/4/2025 10:39 PM	Application extension
System.CodeDom.dll	4/4/2025 10:39 PM	Application extension
System.Collections.Concurrent.dll	4/4/2025 10:39 PM	Application extension

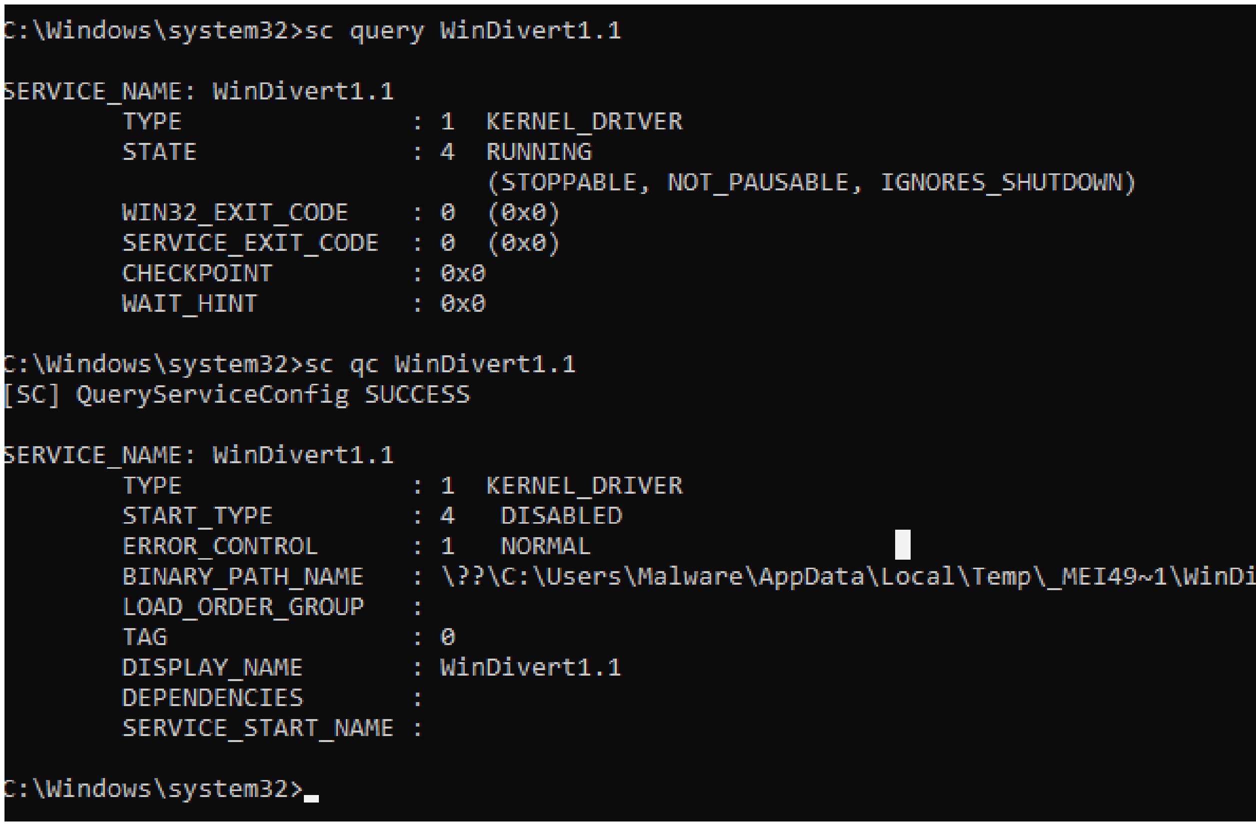
3. Screenshot taken from Process Monitor, you can see that it have a connection to **http://srv760637.hstgr.cloud/**



4 . Screenshot taken from Process Monitor, you can see that it's created the main **AppApi.dll** in the folder `AppData\Local\Temp\.net\Chrome-x64\AV9JX02HGBnwvgGSvHtJPrgR6B2GBbA=`



5. Screenshot taken from the CMD, this can proof that the **Windervert1.1 driver installed and running on the kernel**



6. Screenshot taken from ipinfo.io, this is the malicious IP address:

69.62.119.2

Frankfurt am Main, Hesse, Germany

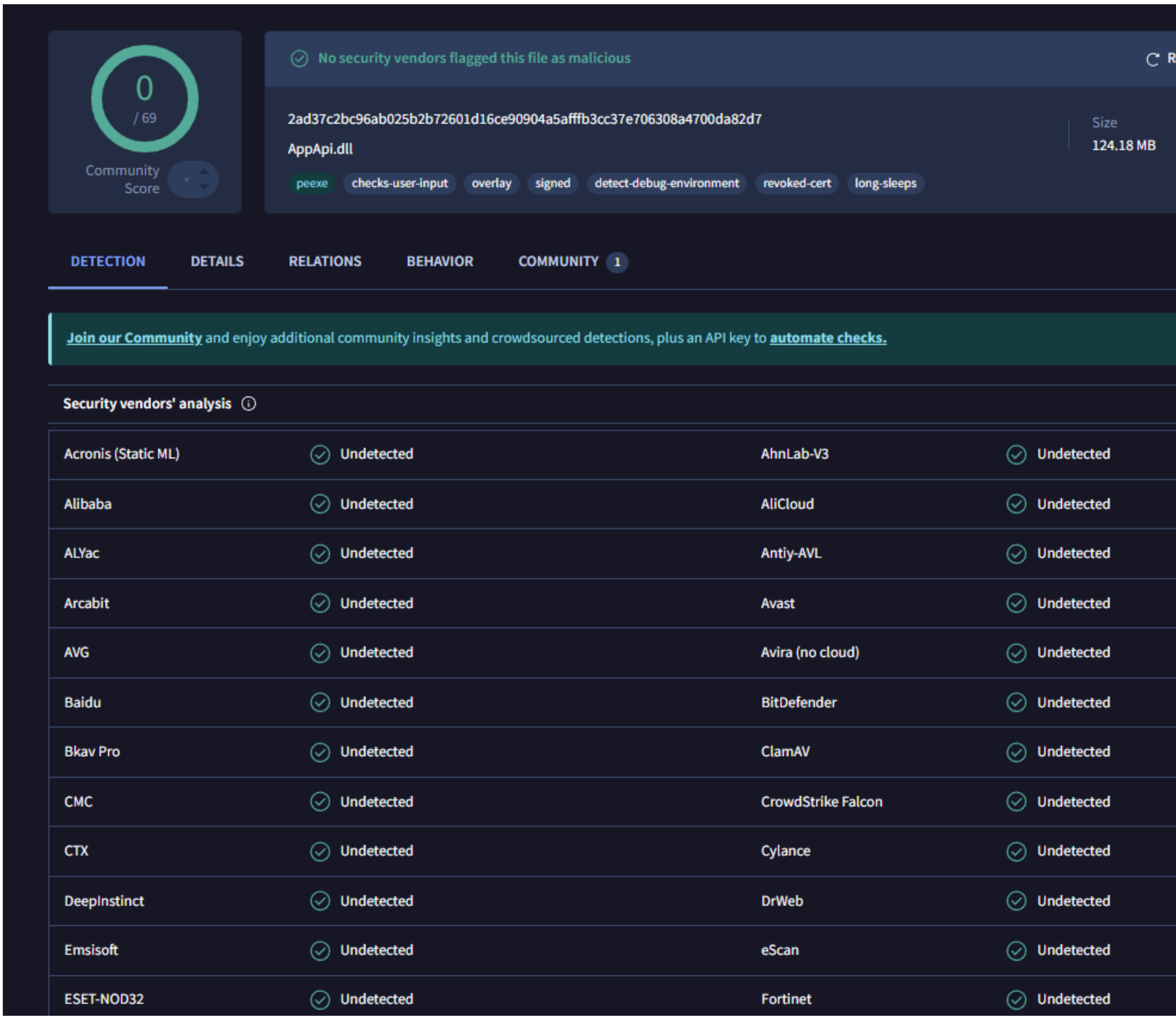
hosting ssh webserver

Need more data or want to access it via API or data downloads? Sign up to get free access

Summary

ASN	AS47583 - Hostinger International Limited
Hostname	srv760637.hstgr.cloud
Range	69.62.116.0/22
Company	Hostinger Operations, UAB
Hosted domains	0
Privacy	True
Anycast	False
ASN type	Hosting
Abuse contact	abuse@hostinger.com

7. Screenshot taken from VirusTotal at March 29, 2025:



Share this report to raise awareness and help others avoid falling for similar traps!

😊